

VIEWPOINT

Fatigue in Air Traffic Operations—Opportunities to Enhance Safety

Mark R. Rosekind, PhD; Erin E. Flynn-Evans, PhD, MPH; Charles A. Czeisler, PhD, MD

Every day in the US, 2.9 million passengers travel on 45 000 flights guided by more than 13 000 air traffic controllers operating in 313 facilities.¹ Air traffic operations involve the safe navigation of aircraft through the National Airspace System via adherence to rules and procedures that maintain separation of aircraft from other aircraft, weather, and hazards. Controlling air traffic is a cognitively demanding, fast-paced, and often intensive job. Air traffic volume can vary greatly day to day and by time of day based on weather, aircraft and airport maintenance issues, and other factors that can lead to congestion at airports, including taxiways, and in flight, thereby greatly increasing task complexity and workload for controllers.

Air traffic controllers confront challenges well-known to those working in 24/7 operational settings, amplified in safety-sensitive environments where errors can mean lives lost and people injured. The newly appointed Administrator of the US Federal Aviation Administration (FAA) requested an independent scientific evaluation of the fatigue risks in current air traffic operations, which we conducted in the first quarter of 2024.² The purpose of the evaluation was to inform the FAA's ongoing efforts to enhance the safety and well-being of the controller workforce and the safety of the aviation system²; here we summarize the key findings and implications of this report with references to relevant medical literature.

Given the safety-sensitive nature of air traffic operations and the requirement for 24/7 operations, concerns about air traffic controller fatigue have been raised for decades.^{2,3} Fatigue related to sleep loss and circadian disruption is known to have significant adverse effects across safety, performance, health, and mood that were highlighted in the report (Appendix D with 135 references).² Some sleep and circadian factors relevant to the work of air traffic controllers include acute sleep loss due to extended time spent awake, chronic sleep loss due to insufficient sleep, circadian timing, sleep inertia, interactions between acute and chronic sleep loss and circadian phase on alertness and performance, medication use, and medical conditions that impact sleepiness.⁴

Based on a review of documents, in-person meetings and interviews, visits to air traffic facilities, and an analysis of data from 747 177 individual reports of work hours and days off from 10 363 controllers during the first 10 weeks of 2024, the panel report focused on strengths of the existing operations, risks, and opportunities to reduce fatigue. The report identified 58 opportunities to address fatigue risks related to the workforce, work requirements, and scheduling practices, and included recommendations in 4 priority areas:

Integrating Prescriptive Policies With Fatigue Risk Management Activities

Although the FAA has pursued fatigue-related activities for more than 40 years (including research, education, regulation, policies, a col-

lective bargaining agreement with the National Air Traffic Controllers Association [NATCA], Fatigue Risk Management System activities, external reviews, and surveys), inconsistencies in how this information has been managed, generated, and distributed were identified as a priority opportunity. Integrating prescriptive policies and regulations and Fatigue Risk Management System activities into an appropriately structured single system would provide a single-source repository of all relevant materials, ensuring consistency across elements and emphasizing the integrated and complementary elements of the system.²

Monitor and Eliminate Scheduling Policy Exceedances

The FAA provided data for all controller shifts with operational duties worked for the first 10 weeks of 2024 (December 31, 2023–March 10, 2024). Sample findings included 470 shifts (0.1%) during this interval that were longer than 10 hours in duration, exceeding FAA Order 7210.3.DD. There were 442 shifts (0.1%) that began with less than 8 hours off between shifts, including 192 shifts (0.04%) that began with less than 4 hours off. There were 562 morning shifts (1.0%) that began with less than 9 hours off after the prior evening shift, and 148 midnight (overnight) shifts (0.44%) preceded by less than 8 hours off duty. Together, these totaled 1622 shifts (0.22%), representing 23 shifts per day, during the first 10 weeks of the year that exceeded current policies and rules and created increased fatigue risk. A second priority opportunity identified by the panel is “to identify and determine specific circumstances around a subset of representative scheduling policy and agreement exceedances and then implement mechanisms to monitor and eliminate them.”² This effort should not involve punitive actions for past circumstances.

Replace the Counterclockwise 2-2-1 Schedule

There is 1 specific schedule that has been examined extensively over 40 years, a counterclockwise rotating 2-2-1 schedule that compresses 5 shifts into 4 days followed by 80 hours off. This schedule features an evening shift followed by an early morning shift after a quick turnaround of only 9 hours off, compressing 2 shifts within 25 hours. The last shift is a midnight shift that typically starts on the same day as the prior day's shift and involves a quick turnaround of 8 hours off between shifts, compressing 2 shifts within 24 hours. This is a relatively unique schedule among 24/7 operations because it is exceptionally challenging to human physiology due to 2 quick turnarounds, with 2 shifts occurring on the same day separated by a short daytime sleep opportunity. The previous shifts in the schedule rotate counterclockwise (earlier) in opposition to the natural rhythm of the circadian clock.

Air traffic controllers on this compressed schedule averaged only 2.4 to 3.5 hours of sleep prior to the midnight shift and have reported operational deviations (6%) and errors (7%), with 51% at-

tributed to fatigue. These controllers report themselves unintentionally falling asleep (77%), experiencing lapses of attention at work (79%), and falling asleep at the wheel while commuting (36%).² In 2014, a National Academies report emphasized the significant risks associated with this schedule, referred to as the “rattler,” including that “it results in severely reduced cognitive performance during the midnight shift because of fatigue.”⁵ This schedule creates known fatigue risks documented by experts, reports, and surveys and yet continues, with a majority of controllers working a single midnight shift (69.3%).² A third priority opportunity identified by the panel is “to develop and implement a strategy to eliminate the counter-clockwise rotating 2-2-1 schedule and replace it with a schedule that meets operational requirements and incorporates sleep and circadian principles.”²

Update Current Policies to Address Fatigue Risks

A fourth priority opportunity is to “develop and implement a strategy to update the current prescriptive policies to address identified fatigue factors, especially to avoid known schedule practices that induce fatigue. Specifically, require sufficient time off-duty (e.g., 10-12 hours) before all shifts, whether controllers are performing operational or non-operational tasks. Also, this off-duty time should account for the circadian timing of the shift, where increased off-duty time may be required before midnight shifts.”² To provide an 8-hour sleep opportunity, an off-duty interval must include 8 hours plus sufficient time for other activities of daily living, such as com-

muting, eating, personal hygiene, and family. This typically equates to a minimum of 10 hours off duty between shifts. Because the off-duty sleep opportunity before and after a midnight shift is affected by circadian misalignment and the evening wake maintenance zone, a minimum of 12 hours off duty prior to and after midnight shifts is recommended to provide sufficient time for sleep. Obtaining sufficient sleep before each work shift is fundamental to ensuring optimal performance and minimizing fatigue risks during work.^{6,7}

The acknowledged air traffic controller staffing shortage can affect fatigue risks through overtime needs, continuous days worked, and scheduling exceedances.⁸ However, even at full staffing, a variety of significant fatigue risks would remain due to the 24/7 demand of air traffic operations.

On April 19, 2024, the FAA Administrator released the panel report and initial updates to fatigue rules that incorporate the opportunities identified in the report, including mandating at least 10 hours off before all shifts and 12 hours off before night shifts.⁹ On July 24, 2024, the FAA Administrator announced an agreement with NATCA that addresses all 4 of the priority opportunities identified by the Scientific Expert Panel, as well as several others.¹⁰ These changes represent meaningful, scientifically based efforts to reduce, mitigate, and eliminate significant fatigue risks in current air traffic operations. The FAA and NATCA are now collaboratively engaged in the complex task of developing the specific policies and actions needed to implement these historic efforts.

ARTICLE INFORMATION

Author Affiliations: Mountain View, California (Rosekind); Fatigue Countermeasures Laboratory, NASA Ames Research Center, Mountain View, California (Flynn-Evans); Division of Sleep Medicine, Harvard Medical School, Boston, Massachusetts (Czeisler); Division of Sleep and Circadian Disorders, Brigham and Women's Hospital, Boston, Massachusetts (Czeisler).

Corresponding Author: Mark R. Rosekind, PhD, 276 Apricot Ln, Mountain View, CA 94040 (nhtsa15@gmail.com).

Published Online: October 28, 2024.
doi:10.1001/jama.2024.21057

Conflict of Interest Disclosures: Dr Rosekind reported receiving personal fees from MBO Partners. Dr Flynn-Evans reported receiving grants from FAA outside the submitted work. Dr Czeisler reported receiving personal fees from Reyes Law Firm, Cloward Trial Lawyers, Haglund Kelley, Adam David Law Firm, Clement Law Firm, Rabb & Rabb, Segal Law Firm, Shaked Law Firm, Zehl Law Firm, Associated Professional Sleep Societies, Klarman Family Foundation, Massachusetts Medical Society, National Council for Mental Wellbeing, National Sleep Foundation, Simpson & Simpson, Tencent Holdings, Teva Pharma Australia, The Armstrong Firm, and Deep; honoraria from Merck Sharpe & Dohme; equity from Signos and Vanda Pharmaceuticals; research/education support to institution from Apnimed, Bryte Foundation, Avadel Pharmaceuticals, Vanda Pharmaceuticals, Harmony Biosciences, Jazz Pharmaceuticals, Stuart F. and Diana L. Quan Charitable Fund, Sleep Number Corporation, Axsome Therapeutics, Zoll, Eisai, Illumination Technology, Takeda, Philips Respironics, Idorsia, Casey Feldman Foundation,

f.lux, Abbaszadeh Foundation, Alexandra Drane, Cephalon, DR Capital, Johnson & Johnson, Neurocare, Optum, Peter Brown and Margaret Hamburg, Regional Home Care, ResMed Foundation, San Francisco Bar Pilots, and WHOOP; and grants from Axsome Therapeutics, Delta Airlines, Jazz Pharmaceuticals, Puget Sound Pilots, CDC Foundation, City of San Francisco, Dayzz Live Well, and Regeneron/Sanofi outside this work.

Funding/Support: The FAA provided financial support through Guidehouse to Dr Rosekind and institutional support to Brigham and Women's Hospital (to Dr Czeisler) for this work.

Role of the Funder/Sponsor: The FAA and Guidehouse had no role in the preparation, review, or approval of the manuscript and decision to submit the manuscript for publication.

Additional Contributions: The Scientific Expert Panel acknowledges valuable contributions from FAA subject matter expert liaisons, the Federal Air Surgeon, Civil Aerospace Medical Institute researchers, and staff and leadership who coordinated facility visits. NATCA senior leadership provided critical information and perspectives. The National Transportation Safety Board Chair and air traffic and fatigue investigators discussed relevant past investigations. The Panel acknowledges important support from former FAA Administrator Michael Huerta, MPA (2023 Safety Review Team report); Jason Sullivan, BS (Brigham and Women's Hospital), for data analysis; and Cassie Hilditch, PhD (San Jose State University); Kelsey Glatfelter, BA, Nataliya Sokolowski, MPP, and Kathleen Davis, BS (Guidehouse), for invaluable management support.

REFERENCES

1. FAA. Air traffic by the numbers. Updated September 9, 2024. Accessed September 26, 2024. https://www.faa.gov/air_traffic/by_the_numbers
2. FAA. Assessing fatigue risk in FAA air traffic operations, a report by Scientific Expert Panel on Air Traffic Controller Safety, Work Hours, and Health. April 19, 2024. Accessed September 26, 2024. https://www.faa.gov/newsroom/media/Fatigue_Report.pdf
3. Gore A Jr, Reid HM, Volkmer HL, et al. Biological clocks and shift work scheduling. In: *Hearings Before the Subcommittee on Investigations and Oversight of the Committee on Science and Technology House of Representatives*. US Government Printing Office, 1983.
4. Duffy JF, Zitting KM, Czeisler CA. The case for addressing operator fatigue. *Rev Hum Factors Ergon*. 2015;10(1):29-78. doi:10.1177/1557234X15573949
5. *The Federal Aviation Administration's Approach for Determining Future Air Traffic Controller Staffing Needs*. National Academies Press, 2014.
6. Hirshkowitz M, Whiton K, Albert SM, et al. National Sleep Foundation's updated sleep duration recommendations. *Sleep Health*. 2015;1(4):233-243.
7. Watson NF, Badr MS, Belenky G, et al. Recommended amount of sleep for a healthy adult. *Sleep*. 2015;38(6):843-844. doi:10.5665/sleep.4716
8. National Airspace System Safety Review Team. Discussion and recommendations to address risk in the National Airspace System. November 2023. Accessed September 26, 2024. https://www.faa.gov/NAS_safety_review_team_report.pdf
9. FAA. Statement from FAA Administrator Mike Whitaker. April 19, 2024. Accessed September 26,

2024. <https://www.faa.gov/newsroom/statement-faa-administrator-mike-whitaker>

10. FAA. FAA and NATCA reach agreement to address controller fatigue by providing more rest between shifts. July 24, 2024. Accessed September

26, 2024. <https://www.faa.gov/newsroom/faq-and-natca-reach-agreement-address-controller-fatigue-providing-more-rest-between>